AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

solution

October 20, 2011

Open Book/Open Notes

In the beginning of your answer sheet, please sign the honor code: I never received nor gave help regarding this exam.

1. (40) The Clohessy-Wiltshire equations describe the orbital perturbation dynamics of a spacecraft in a low Earth orbit, relative to the LVLH rotating coordinate frame at orbital radius R=*6,578km*. The spacecraft mass m=*100kg*. The Clohessy-Wiltshire equations can be approximated by

$$m\Delta\ddot{r} = F_r$$
$$mr\Delta\ddot{\nu} = F_\nu$$
$$mr\Delta\ddot{\lambda} = F_\lambda$$

Here $(\Delta r, \Delta\nu, \Delta\lambda)$ are perturbations of the spacecraft position from the circular orbit, and (F_r, F_ν, F_λ) are propulsive force components expressed in the LVLH rotating coordinate frame. This approximation is often acceptable over a sufficiently small time period $T>0$.

a. Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about *5%*.

b. Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\vec{e}_r km$ and initial velocity vector $\Delta\dot{\vec{r}}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60sec) = 0$ and terminal velocity vector $\Delta\dot{\vec{r}}(60sec) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

2. (30) A spacecraft is launched from Earth to reach the L_2 Lagrange equilibrium. At a point on its trajectory, navigation measurements indicate that the spacecraft position vector is $\vec{r} = -445{,}645{,}6\vec{\imath} + 500\vec{\jmath} + 200\vec{k}\ km$ and the spacecraft velocity vector is $\dot{\vec{r}} = -0.2\vec{\imath} - 0.3\vec{\jmath} + 0.4\vec{k}\ km/sec$, both defined with respect to Earth-Moon barycentric coordinate frame. It is desired to determine a propulsive force vector that transfers the spacecraft from this state to the L_2 Lagrange equilibrium in *7200sec*.

a. What are the linear perturbation equations that describe the spacecraft dynamics with respect to the Earth-Moon coordinate frame whose origin is located at the L_2 Lagrange point? What are the initial conditions and the terminal conditions that define the transfer maneuver?

b. Does this transfer maneuver have a solution? That is, do you know that this transfer maneuver can be accomplished? Give your reasoning. Comments. the problem is asking controllability

c. Once the spacecraft is transferred exactly to the L_2 Lagrange equilibrium, the spacecraft will always remain at this equilibrium, at least in theory. In practice, will the spacecraft always remain at this Lagrange equilibrium? Give your reasoning. unstable inplane out-of-plane

3. (30) A *100kg* spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the restricted 3-body model, answer the following questions.

a. If there is no propulsive force vector, is the spacecraft in equilibrium at this midway point between the Earth and Moon? If it is in equilibrium, prove your result. If it is not in equilibrium, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?

b. What is the constant propulsive force vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your propulsive force vector in the barycentric coordinate frame?